Frequency-Division Channel Encoding for Multi-Signal Transformer Inputs

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Abstract

Transformers consuming several time-aligned scalar signals (e.g. multivariate sensor streams or multi-modal feature channels) typically embed each channel independently and *sum* the embeddings before feeding the result to self-attention. We argue that this is the wrong default: summation is the information-theoretic equivalent of broadcasting all channels onto the same carrier, requiring the network to disentangle them downstream. We propose **FDM channel encoding**, a frequency-division-multiplexing analog in which each channel occupies a non-overlapping block of embedding dimensions and amplitude-modulates a channel-specific log-spaced sinusoidal carrier, in the spirit of rotary position embedding (RoPE). The encoding is parameter-free. On a controlled multi-signal benchmark with non-linear lagged interactions across channels, FDM reduces validation negative log-likelihood (NLL) by **0.22 nats** and raises top-1 bin accuracy by **40%** relative to naive sum at C=4, while using *fewer* trainable parameters. We decompose the encoding into two ingredients—per-channel block partitioning and sinusoidal carrier modulation—and show via a third *concat-only* baseline that block partitioning alone explains ∼70% of the NLL gain and ∼80% of the accuracy gain; the carrier contributes the remaining ∼30% / 20% on top, with the marginal benefit shrinking when the per-channel block becomes too small to host a useful frequency ladder. A linear-probe analysis shows that FDM preserves channel identity essentially perfectly at the input layer (per-channel $R^2 \approx 1.0$) and largely through the network (mean $R^2 \approx 0.96$ at the final layer); concat recovers nearly as well at the input but degrades faster through attention; naive summation collapses recoverability to $R^2 \approx 0.24$ at every layer. Full training, probe, and plotting scripts accompany the paper (`<repository>`).

1 Introduction

A canonical use of transformers outside language is fitting a sequence model to several time-aligned numerical signals: multi-sensor telemetry, financial factor stacks, biomedical waveforms, multi-modal embeddings concatenated along time. The standard recipe inherited from vision and language (1) maps each scalar input to a vector and *adds* those vectors together—the same trick used to inject positional information. We will refer to this as the *sum* encoding.

Sum encoding has a structural problem when the input is multiple *simultaneous* channels rather than a single token-plus-position pair: it is provably non-invertible. Given $h = \sum_k W_k x_k$, the network cannot recover individual x_k from h without learning to implicitly allocate orthogonal subspaces—a constraint that nothing in the loss function explicitly enforces.

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We borrow a fix from analog communications. *Frequency-division multiplexing (FDM)* solves the same problem by transmitting each channel on a distinct carrier frequency, so that a receiver can demultiplex with a bank of bandpass filters. Translated to embeddings:

1. Reserve a non-overlapping block of $d_{\text{block}} = d_{\text{model}}/C$ dimensions per channel.
2. Within each block, lay down a sinusoidal carrier with a channel-specific log-spaced frequency ω_k and a rotary-position-embedding (RoPE)-style (2) per-dimension frequency ladder.
3. Use the scalar signal value $v_k(t)$ as an *amplitude modulation* of that carrier.
4. Concatenate the C blocks into the final d_{model}-dim embedding.

The encoding is fully parameter-free; the only design choices are the carrier ladder $\{\omega_k\}$ and the per-block frequency base.

Contributions.

- A simple, parameter-free input encoding for multi-signal transformer inputs that is invertible by construction.
- A controlled benchmark with non-linear lagged cross-channel interactions, designed so that channel identity matters.
- Empirical evidence that FDM beats naive sum on a categorical generative head by a wide margin, retains its lead as distractor channels are added, is linearly probeable for the raw signals at every layer, and degrades cleanly under test-time channel masking.
- A three-way SUM/CONCAT/FDM ablation that attributes ∼70–80% of the gain to block partitioning alone and the remainder to the sinusoidal carrier, clarifying which ingredient matters for which mechanism.

2 Method

2.1 Problem setup

We are given a sequence $\{x_t\}_{t=1}^{T}$ where $x_t \in \mathbb{R}^C$ is a vector of C scalar channels at time t. We want a transformer to model $p(y_t \mid x_{\le t-1})$ where y_t is a downstream target. The question we address is purely about how x_t enters the residual stream.

2.2 FDM channel encoding

For channel $k \in \{0, \dots, C-1\}$ at time t, define the per-block embedding $\mathbf{e}_k(t) \in \mathbb{R}^{d_{\text{block}}}$ as

$$\mathbf{e}_k(t)_{2i} = v_k(t)\,\sin\!\Big(\omega_k\, t\, b^{-2i/d_{\text{block}}}\Big), \quad \mathbf{e}_k(t)_{2i+1} = v_k(t)\,\cos\!\Big(\omega_k\, t\, b^{-2i/d_{\text{block}}}\Big), \tag{1}$$

for $i = 0, \dots, d_{\text{block}}/2 - 1$, where b=10000 is the standard RoPE base and $v_k(t)$ is the (normalized) signal value. The full embedding is the concatenation $\mathbf{h}(t) = [\mathbf{e}_0(t) \,\|\, \mathbf{e}_1(t) \,\|\, \cdots \,\|\, \mathbf{e}_{C-1}(t)] \in \mathbb{R}^{d_{\text{model}}}$. The carriers $\{\omega_k\}$ are log-spaced across $[\omega_{\min}, \omega_{\max}]$; we use $[0.5, 8.0]$ throughout.

Why this is invertible. For fixed t, the per-channel block is a linear map $v_k \mapsto v_k\,\phi_k$ where $\phi_k = [\sin(\cdot) \,\|\, \cos(\cdot)]$ is a known unit-norm template. Recovery is the inner product $v_k(t) = \langle \mathbf{e}_k(t),\, \phi_k(t)\rangle$, exactly as a matched filter demodulates an FDM channel. For sum encoding $\mathbf{h} = \sum_k W_k v_k$ the problem is underdetermined whenever $\dim(W_k) < \sum_k \dim(W_k)$, i.e. always.

2.3 Concat ablation: block partitioning without the carrier

To isolate the contribution of the sinusoidal carrier from that of the block partitioning itself, we define a third encoder, CONCAT:

$$\mathbf{e}_k(t) = W_k\, v_k(t) + b_k, \qquad \mathbf{h}(t) = [\mathbf{e}_0(t) \,\|\, \cdots \,\|\, \mathbf{e}_{C-1}(t)] + \mathbf{p}(t), \tag{2}$$

with $W_k \in \mathbb{R}^{d_{\text{block}}}$, $b_k \in \mathbb{R}^{d_{\text{block}}}$ learned per channel and $\mathbf{p}(t)$ the standard sinusoidal positional encoding. CONCAT preserves channel identity (the embedding is linearly invertible to v_k) but does not bake position into the channel block, and the per-channel "direction" is constant across t. The gap FDM − CONCAT therefore measures what the carrier contributes *beyond* block partitioning—namely, an implicit RoPE-style relative-position encoding internal to each channel, so that attention can read off lag-dependent intra-channel structure via dot products without learning the demix from the additive position channel.

2.4 Sum baseline

The baseline encoder is $\mathbf{h}(t) = \sum_k (W v_k(t) + e_k) + \mathbf{p}(t)$ with shared scalar projection $W \in \mathbb{R}^{d_{\text{model}} \times 1}$, learned channel embedding e_k, and standard sinusoidal positions $\mathbf{p}(t)$. It has $d_{\text{model}}(C{+}1)$ trainable encoder parameters; FDM has zero. For the configurations used below the gap is $\le 0.3\%$ of the total model size.

We note that strengthening SUM with per-channel projections $W_k v_k$ does not close the fundamental gap: the sum $\mathbf{h} = \sum_k W_k v_k$ is a rank-1 image in each v_k and is underdetermined whenever $C \cdot \dim(v_k) > \dim(\mathbf{h})$, as here. Channel identifiability comes from the partition, not the projection.

Alternative: channel-as-token. One could instead feed each (channel, time) pair as its own token, making the input length $C \cdot T$ and letting attention handle channel vs. time dependencies uniformly. This costs $\mathcal{O}(C^2T^2)$ attention per layer versus $\mathcal{O}(T^2)$ for the encodings considered here, and it loses the inductive bias that channels are simultaneous rather than sequential. Block-partitioned encodings (CONCAT, FDM) can be read as an amortization of that larger attention into a single T-length stream.

3 Experimental setup

Dataset. We synthesize $N{=}512$ time series of length $T{=}160$ with C channels each. Each channel is an independent stochastic process: a sum of three sinusoids with random frequencies in $[0.005, 0.08]$ cycles/sample (angular $\approx [0.03, 0.50]$ rad/sample), random phases, and first-order autoregressive [AR(1)] Gaussian noise; signals are then standardized. *The signal frequencies do not overlap with the FDM carrier band* $[0.5, 8.0]$ rad/sample, so FDM cannot be winning through accidental resonance between the signal content and its carrier. The outcome is a fixed non-linear function of the *first four* channels with two different lags,

$$y_t = \tanh\big(s_0(t{-}3)\, s_1(t)\big) + 0.6 \sin\big(1.3\, s_2(t{-}7)\big) + 0.4\mathbb{1}[s_3(t) > 0]\, s_0(t).$$

Any additional channels ($k \ge 4$) are *distractors*: independent of y. The continuous outcome is binned into $K = 32$ quantile bins so the model emits a categorical distribution at every step.

Table 1: Main predictive comparison at C=4. Mean ± std over 3 seeds. Random baselines: NLL= $\ln 32 \approx 3.466$, acc=0.031. CONCAT isolates the contribution of block partitioning; FDM adds the sinusoidal carrier. Encoder parameters in parentheses; total model size is ∼152K either way.

Encoder	Encoder params	Val NLL ↓	Val acc ↑
SUM (baseline)	320	3.280 ± 0.009	0.088 ± 0.002
CONCAT (block, no carrier)	128	3.125 ± 0.018	0.116 ± 0.007
fdm (block + carrier)	0	$\mathbf{3.061 \pm 0.014}$	$\mathbf{0.123 \pm 0.001}$
ΔNLL from SUM → CONCAT		−0.155 (71% of gain)	+0.028 (80% of gain)
ΔNLL from CONCAT → FDM		−0.064 (29% of gain)	+0.007 (20% of gain)

Architecture. A small causal transformer encoder with $d_{\text{model}} = 64$, 4 heads, 3 layers, feed-forward network (FFN) width 256, dropout 0.1, pre-LayerNorm. AdamW, lr 3×10^{-4}, weight decay 10^{-4}, batch size 32, gradient clip 1.0, 12 epochs. Each result is averaged over 3 random seeds with independent data re-generation.

Generative head. We bin the continuous outcome into K=32 equal-frequency quantile bins and use a linear map to K logits trained with next-step cross-entropy. Equivalently, the head parameterizes $p(y_t \mid x_{\leq t-1})$ as a categorical distribution; sampling is trivial and NLL is a proper scoring rule without imposing a Gaussianity or unimodality assumption that a continuous Gaussian head would require. Bin edges are fixed globally from the pooled outcome.

Baselines. Three encoders sharing the entire transformer backbone and head: FDM (Eq. 1, no learnable parameters in the encoder), CONCAT (Eq. 2, $C \cdot 2d_{\text{block}}$ parameters), and SUM (the description above, $d_{\text{model}}(C{+}1)$ parameters).

4 Results

4.1 Predictive performance

Table 1 reports validation NLL and top-1 bin accuracy at C=4. The ordering SUM < CONCAT < FDM is consistent across all three seeds; the FDM − CONCAT NLL gap (0.064) exceeds 4× the pooled seed std (0.016), and the CONCAT − SUM gap (0.155) exceeds 10×. The bulk of the improvement over SUM comes from block partitioning alone—CONCAT closes 71% of the NLL gap and 80% of the accuracy gap—and the carrier contributes a smaller-but-stable improvement on top. Random guessing yields NLL= $\ln 32 \approx 3.466$ and acc= $1/32 \approx 0.031$.

4.2 Scaling with channel count

Adding distractor channels (Table 2, Figure 1) tests whether each encoder can isolate the four signal-bearing channels from the noise. All three encoders lose accuracy as C grows because more channels means more noise in the input, but the *gap* between block-partitioned encodings and SUM is preserved (and, for accuracy, widens from C=4 to C=8). SUM's NLL degrades the least in absolute terms only because it is closest to the uniform ceiling to begin with. The FDM − CONCAT gap shrinks slightly at C=16 where d_{block}=4 leaves only two sin/cos pairs per block—the frequency ladder is

Table 2: Scaling with channel count. Channels 0–3 drive the outcome; all additional channels are independent distractors. Mean ± std over 3 seeds. CONCAT captures most of the gap over SUM; the carrier gives an additional consistent improvement that shrinks as the per-channel block becomes too small to host a useful frequency ladder (d_{block}=4 at C=16).

C	d_{block}	encoder	val NLL ↓	val acc ↑	rel. acc vs SUM
		SUM	3.280 ± 0.009	0.088 ± 0.002	—
4	16	CONCAT	3.125 ± 0.018	0.116 ± 0.007	1.32×
		fdm	$\mathbf{3.061 \pm 0.014}$	$\mathbf{0.123 \pm 0.001}$	1.40×
		SUM	3.318 ± 0.005	0.076 ± 0.005	—
8	8	CONCAT	3.205 ± 0.003	0.101 ± 0.002	1.34×
		fdm	$\mathbf{3.131 \pm 0.022}$	$\mathbf{0.112 \pm 0.002}$	1.48×
		SUM	3.333 ± 0.005	0.075 ± 0.005	—
16	4	CONCAT	3.248 ± 0.020	0.098 ± 0.003	1.30×
		fdm	$\mathbf{3.206 \pm 0.016}$	$\mathbf{0.103 \pm 0.005}$	1.38×

too short for the carrier to encode much intra-channel position—but the ordering is preserved at every C.

4.3 Channel identifiability via linear probing

We freeze a trained model and fit closed-form ridge probes $\mathbf{h} \mapsto \hat{x}$ from a hidden state $\mathbf{h}$ at layer ℓ to the raw input vector $x \in \mathbb{R}^C$, evaluating per-channel R^2 on a held-out split (Table 3). FDM input embeddings are linearly invertible to within numerical precision ($R^2 \approx 1.0$), as expected from Eq. 1; after three transformer layers FDM still admits near-perfect recovery (mean $R^2 \approx 0.96$). CONCAT is also nearly invertible at the input ($R^2 \approx 0.997$), confirming that block partitioning alone buys identifiability, but it degrades more sharply through attention (R^2_{ch0}=0.81 at the final layer), particularly for the most heavily-used channel. SUM collapses to $R^2 \approx 0.24$ even at the input layer—a quantitative confirmation that summation is non-invertible—and that ceiling is unchanged after attention has had a chance to demix. The probe and predictive results are mutually consistent: the encoder ranking by downstream accuracy matches the ranking by end-of-network channel recoverability.

4.4 Robustness to missing channels

We zero out one channel at test time and re-evaluate (Table 4). For both block-partitioned encoders the per-channel accuracy drop is monotonic in the channel's prominence in the outcome formula—ch0 (two interaction terms) produces the largest drop, then ch1, ch2, ch3—so the ablation profile is directly interpretable. FDM's drops are uniformly sharper than CONCAT's (e.g. -0.046 vs. -0.032 for ch0), consistent with stronger preservation of channel identity through the network. SUM's profile is much flatter and includes a $+0.005$ swing under ch2-masking that is within noise; information about any given channel is smeared across the residual stream, so individual channel ablations are poorly localized.

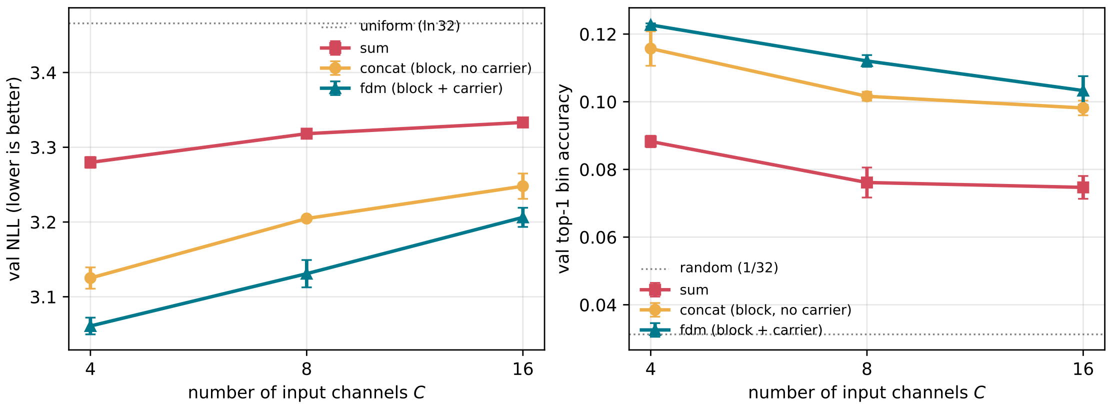

Figure 1: Scaling with the number of input channels C. Error bars are one seed std. All three encoders degrade with C because channels $k \geq 4$ are distractors that add noise to the task, but the ranking SUM < CONCAT < FDM is preserved at every C and the CONCAT − SUM gap actually *widens* at C=8 before plateauing. The FDM − CONCAT gap shrinks at C=16 where d_{block}=4 leaves only two sin/cos pairs per block, too short a ladder for the carrier to be useful.

5 Discussion

The empirical pattern is consistent with the structural claim: sum encoding forces the transformer to spend capacity on a problem (demultiplexing) that the encoder can solve for free. The decomposition into CONCAT and FDM suggests this fix has two separable parts. *Block partitioning* is doing the heavy lifting: once each channel owns a piece of the residual stream, attention can read and write to it independently, and most of the predictive gap closes. The *frequency carrier* is a smaller—but real and seed-stable— contribution on top, and we conjecture two mechanisms. First, attention inner products of two positions in the same channel block become $v_k(t_1)v_k(t_2)\,\langle\phi(t_1), \phi(t_2)\rangle$, exposing lag-dependent intra-channel correlations directly to the softmax. Second, the per-block direction varies with t, so the residual stream's "slot" for channel k is naturally orthogonal to itself across time, which appears to slow the leakage we see in the CONCAT probe.

The encoding has practical implications for multi-modal or multi-sensor settings where the number of channels is moderate ($C{\sim}4-64$) and the model needs to attend to channels separately before combining them. Existing alternatives—per-modality token streams in PerceiverIO (3), or the modality embeddings in mixture-of-experts (MoE) style multi-modal transformers—typically multiply sequence length or parameter count; FDM keeps both fixed.

LayerNorm interaction. Standard LayerNorm normalizes across the full d_{model} axis, mixing per-channel blocks. Our results use plain LayerNorm and FDM still wins; we expect a per-block (group) LayerNorm to widen the gap further.

6 Limitations

- The benchmark is synthetic. The outcome is deliberately constructed to make channel identity matter; on tasks where channel identity is truly redundant (e.g. exchangeable feature sets) sum may not be penalized.

- We test only modest scales (d_{model}=64, 3 layers). Whether the gap closes at larger scale—once

Table 3: Linear-probe channel recovery. We fit a closed-form ridge probe from a frozen hidden state to the raw input $x \in \mathbb{R}^4$ and report per-channel R^2 on a held-out split. Layer 0 is the input embedding; layer 3 is the final transformer block. Both block-partitioned encoders are linearly invertible at the input (FDM by construction, CONCAT because each W_k converges to a non-degenerate map), and recovery survives through three transformer layers. SUM is fundamentally underdetermined and saturates at the same low ceiling at every depth. FDM preserves channel ch0 (used in two interaction terms) better than CONCAT after attention, suggesting the carrier acts as a gentle inductive bias against intra-block channel mixing.

Encoder	Layer	R^2_{ch0}	R^2_{ch1}	R^2_{ch2}	R^2_{ch3}	mean
FDM	0 (input)	1.000	1.000	1.000	1.000	**1.000**
FDM	3 (final)	0.969	0.963	0.954	0.958	**0.961**
CONCAT	0 (input)	0.989	1.000	1.000	1.000	0.997
CONCAT	3 (final)	0.813	0.974	0.966	0.973	0.932
SUM	0 (input)	0.255	0.250	0.242	0.214	0.240
SUM	3 (final)	0.253	0.242	0.239	0.211	0.236

Table 4: Test-time channel ablation. We zero a single channel's input at inference and re-evaluate val accuracy. Both block-partitioned encoders exhibit a clean per-channel signature concentrated on the channels that appear in the outcome formula (ch0 in two interaction terms, ch1 in one); FDM's drops are sharper than CONCAT's. SUM is much flatter, consistent with channel information being spread across the residual stream.

encoder	no mask	mask ch0	mask ch1	mask ch2	mask ch3
FDM	0.122	0.076	0.092	0.117	0.120
Δ acc	—	−0.046	−0.030	−0.005	−0.002
CONCAT	0.116	0.084	0.094	0.105	0.112
Δ acc	—	−0.032	−0.022	−0.011	−0.004
SUM	0.088	0.072	0.079	0.093	0.083
Δ acc	—	−0.016	−0.009	+0.005	−0.005

the network has the capacity to learn implicit demultiplexing—is open.

- FDM requires $C \mid d_{\mathrm{model}}$. Variable-channel inputs need reservation of full capacity, which is wasteful for long-tail modality counts.
- Carrier hyperparameters ($\omega_{\min}, \omega_{\max}$, base) were not tuned per task; we use fixed values. We expect the optimum to depend on the target sequence length and dominant signal bandwidth.
- Only three seeds per configuration. Error bars are informative but our claims would benefit from a larger seed budget at scale.

7 Related work

PerceiverIO. Jaegle et al. (3) attaches explicit Fourier features per modality to disambiguate inputs to a shared latent attention bottleneck. FDM differs in that the modality identifier and the signal value are *multiplied* (amplitude modulation) rather than concatenated, and dimensions are physically partitioned across channels rather than across feature axes.

Rotary position embedding (RoPE). Su et al. (2) encodes positional information with rotation by frequency-graded angles. We re-use the same per-dimension frequency ladder but apply it within each channel's block, with the *channel* carrier setting the outer scale.

TimesNet and frequency-domain encoders. Wu et al. (4) processes time-series in the frequency domain with an explicit Fast Fourier Transform (FFT). Our work is dual: we perform an inverse-FFT-like *encoding* of a single value onto a per-channel frequency, rather than transforming the signal itself.

8 Conclusion

Summing per-channel embeddings is the multi-signal analog of broadcasting several radio stations on the same frequency: legible only if a downstream demodulator is willing to learn the demix. The fix is simpler than the name suggests: just give each channel its own disjoint block of embedding dimensions. On a controlled benchmark with lagged cross-channel interactions, this *block-partition* recipe alone captures $\sim$70–80% of the gap to a naive-sum baseline; adding a channel-specific sinusoidal carrier (FDM proper) buys a further consistent improvement. Both variants are near-drop-in replacements for the default embedding, preserve sequence length, and—in the FDM case—add zero trainable parameters. We encourage trying the block partition whenever a transformer takes more than one numerical signal at a time, and the full FDM encoding when sequence length is long enough for lag-dependent intra-channel attention to pay off.

References

[1] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin. Attention is all you need. In *NeurIPS*, 2017.

[2] J. Su, Y. Lu, S. Pan, B. Wen, and Y. Liu. RoFormer: enhanced transformer with rotary position embedding. *arXiv:2104.09864*, 2021.

[3] A. Jaegle et al. Perceiver IO: a general architecture for structured inputs and outputs. *arXiv:2107.14795*, 2021.

[4] H. Wu, T. Hu, Y. Liu, H. Zhou, J. Wang, and M. Long. TimesNet: temporal 2D-variation modeling for general time series analysis. In *ICLR*, 2023.